Physiology of Blood: Composition, Erythropoiesis, & Blood Groups

1. Blood is a dynamic fluid connective tissue. When anti-coagulated blood is centrifuged in a hematocrit tube, the packed cell volume representing the percentage of total blood volume occupied by erythrocytes is typically:
a. 55% to 60%
b. 10% to 15%
c. 75% to 80%
d. 40% to 45% (Hematocrit)
e. 5% exactly
2. The transparent, straw-colored supernatant fluid remaining at the top of a hematocrit tube after centrifugation contains water, electrolytes, and proteins, and is formally termed:
a. Serum
b. Lymph
c. Interstitial fluid
d. Plasma
e. Chyle
3. Serum is physiologically distinct from plasma. Under precise laboratory evaluation, serum represents plasma that completely lacks which major component?
a. Albumin
b. Immunoglobulins
c. Fibrinogen and other clotting factors
d. Sodium and potassium ions
e. Glucose molecules
4. Plasma proteins exert a crucial osmotic force across the capillary membrane that prevents fluid from leaking excessively into the interstitial spaces. This force is the:
a. Hydrostatic pressure
b. Interstitial fluid pressure
c. Colloid osmotic (oncotic) pressure
d. Intracellular turgor
e. Dynamic blood pressure
5. Which specific plasma protein is the most abundant by weight and is primarily responsible for generating roughly 80% of the total plasma colloid osmotic pressure?
a. Fibrinogen
b. Alpha-1 antitrypsin
c. Prothrombin
d. Albumin
e. Gamma globulin
6. While albumin and fibrinogen are synthesized exclusively by the liver, a major fraction of the globulins (specifically gamma globulins) is synthesized by:
a. Plasma cells (derived from B lymphocytes in lymphoid tissues)
b. Hepatic sinusoidal endothelial cells
c. Renal juxtaglomerular cells
d. Pulmonary alveolar Type II cells
e. Splenic red pulp cord cells

7. Erythrocytes (red blood cells) possess a highly specialized morphology optimized for gas transport. In a healthy adult, a mature erythrocyte is structurally characterized as a:
a. Non-nucleated, biconcave disc
b. Spherical nucleated cell rich in mitochondria
c. Biconvex cell containing a dense Golgi apparatus
d. Multilobed amoeboid cell fragment
e. Flat squamous plate lacking cytoplasm
8. The average lifespan of a mature erythrocyte circulating within the high-shear environment of the human vascular system is approximately:
a. 12 hours
b. 10 days
c. 30 days
d. 365 days
e. 120 days
9. Because mature human erythrocytes completely lack a nucleus, ribosomes, and mitochondria, they are physiologically incapable of replicating or performing oxidative phosphorylation, relying instead for ATP production on:
a. Beta-oxidation of fatty acids
b. The Krebs cycle purely
c. Anaerobic glycolysis (Embden-Meyerhof pathway)
d. The malate-aspartate shuttle
e. Direct photo-activation
10. In a normal, healthy adult, the primary site of red blood cell production (erythropoiesis) across all skeletal segments is the:
a. Spleen
b. Liver
c. Yellow bone marrow
d. Lymph nodes
e. Red bone marrow
11. Let's trace the chronological shifts in embryonic sites of erythropoiesis. During the first few weeks of early embryonic life, primitive, nucleated red blood cells are produced exclusively in the:
a. Yolk sac
b. Fetal liver
c. Thymus
d. Spleen
e. Marrow cavities
12. During the second trimester of gestation, the yolk sac regresses, and the primary organ responsible for the bulk of fetal erythropoiesis becomes the:
a. Fetal kidney
b. Fetal spleen
c. Thymus gland
d. Fetal liver
e. Long bone cavities
13. From the third trimester of fetal life, through birth, and continuing into adulthood, erythropoiesis shifts permanently to the red bone marrow. However, after the age of 20, the marrow of which bones becomes fatty and ceases red cell production?

a. Shafts of the long bones (such as the femur and humerus)
b. Vertebrae
c. Sternum
d. Ribs
e. Iliac crest bones

14. All cellular elements of the blood originate from a single, shared precursor stem cell located within the bone marrow, known as the:
a. Reticulocyte
b. Colony-forming unit-erythroid (CFU-E)
c. Megakaryocyte
d. Pluripotent hematopoietic stem cell (PHSC)
e. Proerythroblast

15. As the PHSC differentiates down the red blood cell lineage, it gives rise to a committed stem cell that produces large clusters of erythroid colonies, designated as the:
a. Granulocyte-monocyte progenitor (CFU-GM)
b. Megakaryocytic progenitor (CFU-Meg)
c. Colony-forming unit-erythroid (CFU-E)
d. Burst-forming unit-erythroid (BFU-E)
e. Lymphoid stem cell

16. The very first structurally identifiable precursor cell of the erythroid series visible under standard microscopic evaluation of bone marrow aspirates is the:
a. Reticulocyte
b. Basophilic erythroblast
c. Proerythroblast
d. Polychromatophilic erythroblast
e. Orthochromatic erythroblast

17. During the basophilic erythroblast stage, the cell accumulates a massive concentration of which cellular organelles to begin the synthesis of hemoglobin?
a. Smooth endoplasmic reticulum
b. Ribosomes
c. Mitochondria
d. Peroxisomes
e. Lysosomes

18. As hemoglobin accumulates, the cytoplasm stains with both basic and acidic dyes. In the subsequent stage, the nucleus condenses to a small size, and the cell is termed a:
a. Polychromatophilic erythroblast
b. Proerythroblast
c. Basophilic erythroblast
d. Orthochromatic erythroblast (normoblast)
e. Reticulocyte

19. At the end of the orthochromatic erythroblast stage, the condensed nucleus is extruded from the cell. The resulting non-nucleated cell contains a faint network of residual organelle structures, characterizing it as a:
a. Reticulocyte
b. Mature erythrocyte
c. Promegakaryocyte

d. Basophilic erythroblast
e. Monocyte

20. The residual network visible inside a reticulocyte under supravital staining (such as new methylene blue) is composed of remnants of:

a. Ribosomes, RNA, and endoplasmic reticulum
b. The nuclear envelope
c. Golgi complexes
d. Mitochondria crests
e. Centriole pairs

21. Reticulocytes pass from the bone marrow into the circulating blood by squeezing through the endothelial pores of the marrow capillaries, a physical migration process known as:

a. Diapedesis
b. Chemotaxis
c. Opsonization
d. Phagocytosis
e. Pinocytosis

22. Under normal physiological conditions, reticulocytes circulate in the blood for approximately 1 to 2 days before transforming into mature erythrocytes. The normal reticulocyte count as a percentage of total RBCs is about:

a. 10% to 15%
b. 20% to 25%
c. 0.5% to 1.5%
d. 0% exactly
e. 5% to 8%

23. Total red blood cell mass in the vascular system is regulated within a narrow physiological range. The primary fundamental stimulus that triggers an acceleration of erythropoiesis is:

a. An increase in systemic blood pressure
b. An excess of plasma proteins
c. High dietary intake of iron salts
d. Tissue hypoxia (decreased oxygen delivery to tissues)
e. A rise in blood glucose levels

24. Tissue hypoxia does not stimulate the bone marrow directly. Instead, it triggers the synthesis and release of a circulating glycoprotein hormone called:

a. Thrombopoietin
b. Interleukin-3
c. Growth hormone
d. Erythropoietin (EPO)
e. Cortisol

25. Under normal physiological conditions, approximately 90% of all systemic erythropoietin is synthesized and released by which organ?

a. Liver
b. Spleen
c. Kidneys
d. Red bone marrow
e. Lung parenchyma

26. The remaining 10% of systemic erythropoietin production is mediated primarily by the:
a. Spleen
b. Pancreas
c. Brainstem
d. Adrenal cortex
e. Liver
27. Within the kidneys, erythropoietin is synthesized primarily by which specialized cell population in response to local hypoxia?
a. Interstitial cells (or peritubular capillary endothelial cells)
b. Podocytes of the glomerulus
c. Proximal convoluted tubule epithelial cells
d. Macula densa cells
e. Cortical collecting duct cells
28. Following a sudden hypoxic insult (such as moving to a high-altitude environment), erythropoietin levels in the blood rise rapidly, reaching peak concentrations within:
a. Erythropoietin levels peak within 24 hours
b. 5 minutes
c. 5 days
d. 3 weeks
e. 2 months
29. Although erythropoietin levels peak within 24 hours of a hypoxic stimulus, newly generated mature erythrocytes do not appear in significant numbers in the circulating blood until:
a. 5 days later
b. 10 minutes later
c. 2 hours later
d. 30 days later
e. 6 months later
30. Erythropoietin accelerates erythropoiesis primarily by stimulating the proerythroblast stage and driving the committed:
a. Colony-forming units-erythroid (CFU-E) to proliferate and differentiate
b. Pluripotent stem cells to die
c. Megakaryocytes to split into platelets
d. Reticulocytes to retain their nuclei
e. Granulocytes to release histamines
31. In chronic renal failure, destruction of the functional kidney parenchyma leads to a severe drop in erythropoietin production, resulting in a profound:
a. Polycythemia
b. Leukocytosis
c. Thrombocytosis
d. Anemia (renal anemia)
e. Eofinophilia
32. Nutritional factors are critical for proper erythropoiesis. Two vitamins are essential for the final maturation of erythrocytes because they are required for the synthesis of thymidine triphosphate, a vital building block of:
a. Cell wall lipids

b. Hemoglobin protein chains
c. Vitamin K factors
d. Carbonic anhydrase enzymes
e. Deoxyribonucleic acid (DNA)

33. A deficiency of either Vitamin B12 (cyanocobalamin) or Folic acid causes a failure of normal DNA replication and nuclear maturation, resulting in the formation of abnormally large, fragile, oval red blood cells called:

a. Microcytes
b. Spherocytes
c. Acanthocytes
d. Schistocytes
e. Megaloblasts (and mature macrocytes)

34. The megaloblastic cells have weak, irregular membranes and are highly fragile. Consequently, their lifespan in the circulation is shortened to one-half or one-third of normal, causing a:

a. Macrocytic (megaloblastic) anemia
b. Hemolytic anemia
c. Microcytic hypochromic anemia
d. Aplastic anemia
e. Sickle cell crisis

35. Absorption of Vitamin B12 from the gastrointestinal tract requires a specialized glycoprotein secreted by the parietal cells of the gastric glands, known as:

a. Pepsinogen
b. Hydrochloric acid
c. Secretin
d. Bile salts
e. Gastric intrinsic factor

36. Gastric intrinsic factor binds to Vitamin B12, protecting it from digestion and facilitating its active absorption in which specific segment of the small intestine?

a. Duodenum
b. Jejunum
c. Cecum
d. Ascending colon
e. Ileum

37. Atrophy of the gastric mucosa or autoimmune destruction of parietal cells leads to a failure of intrinsic factor secretion and a subsequent deficiency of Vitamin B12 absorption, causing:

a. Pernicious anemia
b. Iron deficiency anemia
c. Thalassemia
d. Sickle cell anemia
e. Aplastic anemia

38. Once absorbed from the ileum, Vitamin B12 is transported via the blood to be stored in large quantities within which primary storage organ?

a. Kidneys
b. Red bone marrow
c. Spleen
d. Skeletal muscle

e. Liver

39. Hemoglobin synthesis requires an adequate supply of iron. In the human body, the largest fraction (approximately 65%) of total body iron is located within:

a. Myoglobin molecules
b. Cytochrome enzymes
c. Ferritin storage pools
d. Transferrin transport proteins
e. Hemoglobin inside circulating erythrocytes

40. When iron is absorbed from the small intestine, it immediately combines in the blood plasma with a beta globulin called apotransferrin to form a transport complex named:

a. Transferrin
b. Ferritin
c. Hemosiderin
d. Ceruloplasmin
e. Haptoglobin

41. Inside cytoplasm storage cells (such as hepatocytes and bone marrow macrophages), iron combines with a protein called apoferritin to be stored safely as:

a. Transferrin
b. Ferritin
c. Hemosiderin
d. Hemoglobin
e. Bilirubin

42. If the total quantity of iron in the body exceeds the storage capacity of apoferritin, the excess iron deposits as large, insoluble clusters within the cells, visible microscopically as:

a. Ferritin beads
b. Heinz bodies
c. Howell-Jolly bodies
d. Cabot rings
e. Hemosiderin

43. A deficiency of dietary iron intake or chronic blood loss prevents the synthesis of adequate amounts of hemoglobin, leading to the formation of small, pale red blood cells, a condition classified as:

a. Macrocytic hyperchromic anemia
b. Normocytic normochromic anemia
c. Aplastic anemia
d. Hemolytic jaundice
e. Microcytic, hypochromic anemia (iron deficiency anemia)

44. Let's shift to blood grouping systems. The membranes of human erythrocytes contain a wide variety of inherited antigenic proteins and lipopolysaccharides, known physiologically as:

a. Agglutinins
b. Opsonins
c. Antibodies
d. Agglutinogens (or antigens)
e. Hemolysins

45. The blood plasma of an individual contains specific gamma globulin antibodies that react against the agglutinogens absent from their own red blood cells. This individual profile contains:
a. Agglutinogens
b. Complement factors
c. Major histocompatibility complexes
d. Perforins
e. Agglutinins (or isoantibodies)
46. In the ABO blood grouping system, two primary agglutinogens are present on the surface of erythrocytes. These are:
a. Antigen C and Antigen D
b. Type A and Type B agglutinogens
c. Antigen M and Antigen N
d. Rh factor factor keys
e. Kell and Duffy markers
47. An individual whose red blood cell membranes express *only* the Type A agglutinogen belongs to which specific ABO blood group category?
a. Blood Group B
b. Blood Group AB
c. Blood Group O
d. Blood Group D
e. Blood Group A
48. An individual whose erythrocyte membranes express *both* Type A and Type B agglutinogens simultaneously belongs to Blood Group AB. Which agglutinins are present in their plasma?
a. Neither Anti-A nor Anti-B agglutinins (plasma lacks both)
b. Anti-A agglutinin only
c. Anti-B agglutinin only
d. Both Anti-A and Anti-B agglutinins
e. Anti-D exclusively
49. Blood Group O is characterized by the absolute absence of both Type A and Type B agglutinogens on the erythrocyte membranes. Consequently, the plasma of a Group O individual contains:
a. No antibodies at all
b. Anti-A antibodies only
c. Anti-B antibodies only
d. Anti-Rh antibodies purely
e. Both Anti-A and Anti-B agglutinins
50. Agglutinins are typically not present in the blood plasma at birth. They begin to develop in the infant's plasma around 2 to 8 months after birth, a process stimulated primarily by:
a. Exposure to Type A and Type B antigens present in common foods and intestinal bacteria
b. The breakdown of maternal milk proteins
c. The maturation of the fetal thymus gland
d. Inhalation of airborne fungal spores
e. Direct genetic transcription in the marrow without external stimuli
51. The anti-A and anti-B agglutinins circulating in the plasma are large immunoglobulins. Which specific structural immunoglobulin class do they belong to?

a. IgG
b. IgA
c. IgE
d. IgD
e. IgM (predominantly, which do not cross the placenta)

52. When mismatched blood is introduced during a blood transfusion, the recipient's plasma agglutinins bind to the donor's red blood cells, causing them to clump together, a process called:

a. Coagulation
b. Agglutination
c. Hemolysis purely
d. Precipitation
e. Flocculation

53. Following agglutination, the clumped red blood cells are broken down physically or destroyed by phagocytic cells, or undergo immediate intravascular rupture mediated by complement activation, a process termed:

a. Hemolysis
b. Agglutination
c. Opsonization
d. Erythropoiesis
e. Aggregation

54. The major immediate hazard of a mismatched blood transfusion is acute intravascular hemolysis. The rapid destruction of donor red cells releases massive amounts of free hemoglobin into the plasma, which can cause:

a. Hyperthyroidism
b. Acute renal shutdown (renal failure due to tubular damage)
c. Profuse salivation
d. Severe alkalosis
e. Aplastic anemia

55. Let's move to the Rh blood types. The Rh system is based on several distinct antigens on the erythrocyte membrane. The most immunogenic and clinically significant Rh antigen is:

a. Antigen C
b. Antigen c
c. Antigen D
d. Antigen E
e. Antigen e

56. An individual whose red blood cells express the Antigen D protein on their surface is classified in clinical practice as:

a. Rh-negative
b. ABO-incompatible
c. Universal donor
d. Secretor positive
e. Rh-positive

57. Unlike the ABO system, an Rh-negative individual does not possess preformed anti-D agglutinins in their plasma under normal conditions. Anti-D agglutinins develop only when the individual is exposed to Rh-positive blood via:

a. Ingestion of raw meat

b. Transfusion of Rh-positive blood or pregnancy with an Rh-positive fetus
c. Universal vaccination protocols
d. Inhalation of atmospheric dust
e. Contact with maternal saliva

58. Once an Rh-negative person is sensitized to the D antigen, the anti-D antibodies generated are smaller and can track through tissue barriers. These antibodies belong to which class?

a. IgM
b. IgA
c. IgG (which can cross the placental barrier easily)
d. IgE
e. IgD

59. Erythroblastosis fetalis (hemolytic disease of the newborn) is a severe clinical condition that can occur when an Rh-negative mother carries a fetus who is:

a. Rh-negative
b. Blood Group O
c. Rh-positive (inherited from an Rh-positive father)
d. AB-negative
e. Secretor negative

60. In erythroblastosis fetalis, fetal red blood cells escape into the maternal circulation during delivery, causing the mother to synthesize anti-Rh (anti-D) antibodies. During a subsequent pregnancy, these maternal IgG antibodies:

a. Are completely destroyed by the placenta
b. Cross the placenta into the fetal blood, causing agglutination and hemolysis of fetal erythrocytes
c. Accelerate fetal lung maturation
d. Stimulate fetal erythropoietin synthesis in the liver
e. Transform into anti-A antibodies

61. Rapid hemolysis of fetal red cells inside an erythroblastotic newborn releases massive amounts of hemoglobin, which is metabolized into a lipid-soluble pigment, resulting in severe:

a. Cyanosis
b. Albinism
c. Melanosis
d. Erythema
e. Jaundice (icterus gravis neonatorum)

62. Because the fetus loses its circulating red cells rapidly to maternal antibody destruction, the fetal hematopoietic organs adapt by releasing large numbers of immature, nucleated red cell precursors into the blood, named:

a. Reticulocytes only
b. Erythroblasts (giving the disease its name)
c. Megakaryocytes
d. Monoblasts
e. Myelocytes

63. If the concentration of lipid-soluble bilirubin rises excessively in the newborn's blood, it can cross the immature blood-brain barrier to deposit in and destroy the basal ganglia, causing permanent motor and mental deficits, a condition termed:

a. Horner's syndrome

b. Hydrocephalus
c. Parinaud's syndrome
d. Bell's palsy
e. Kernicterus

64. To prevent maternal sensitization to the D antigen, an Rh-negative mother is treated prophylactically with anti-D gamma globulin antibodies (Rh0GAM) at 28 weeks of gestation and within 72 hours of delivery. This passive antibody therapy acts to:
a. Destroy any fetal Rh-positive cells in the maternal blood before the mother's immune system can respond
b. Supplement the fetal hemoglobin levels
c. Suppress the fetal bone marrow erythropoiesis
d. Neutralize the maternal ABO antibodies
e. Stimulate maternal IgM synthesis

65. Let's revisit erythrocyte functions. Erythrocytes contain a massive concentration of an enzyme that catalyzes the reversible reaction between carbon dioxide and water, enabling the blood to transport massive amounts of CO2 as bicarbonate. This enzyme is:
a. Alkaline phosphatase
b. Cytochrome P450
c. Lactate dehydrogenase
d. Hexokinase
e. Carbonic anhydrase

66. Hemoglobin acts as an excellent acid-base buffer within the circulating blood, responsible for a large fraction of the total buffering power of whole blood. This is due to its high concentration of:
a. Sodium ions
b. Glucose chains
c. Amino acid residues (such as histidine) that can bind or release hydrogen ions
d. Lipid bilayers
e. Phosphate crystals

67. In a healthy adult male, the average number of erythrocytes per cubic millimeter (microliter) of blood typically ranges around:
a. 1 million
b. 2.5 million
c. 8.5 million
d. 500,000 exactly
e. 5.2 million (plus or minus 300,000)

68. In a healthy adult female, the average normal red blood cell count per cubic millimeter of blood is approximately:
a. 4.7 million (plus or minus 300,000)
b. 3.2 million
c. 6.5 million
d. 1.2 million
e. 9.0 million

69. Each mature erythrocyte can pack up to 34 grams of hemoglobin per 100 milliliters of cells. In a normal man, the concentration of hemoglobin per 100 milliliters of whole blood averages:
a. 5 grams

b. 10 grams
c. 15 grams (typically 14 to 16 g/dL)
d. 25 grams
e. 34 grams

70. Each individual gram of pure hemoglobin is physiologically capable of binding chemically with a maximum volume of oxygen equal to:

a. 0.5 mL
b. 5.0 mL
c. 10.0 mL
d. 22.4 mL
e. 1.34 mL of oxygen

71. Polycythemia vera is a clonal stem cell disorder characterized by an abnormal, uncontrolled proliferation of the erythroid series. This leads to a massive rise in the hematocrit and a marked increase in blood:

a. Alkalinity
b. Volume velocity
c. Viscosity (making blood flow sluggish)
d. Velocity through capillaries
e. Temperature

72. Secondary polycythemia occurs as a physiological adaptation to chronic tissue hypoxia. It is commonly observed in individuals who live at:

a. Deep sea level trenches
b. Highly elevated altitudes (above 10,000 to 14,000 feet)
c. Controlled hyperbaric chambers
d. Humid tropical coastlines
e. Dark caves

73. During hemoglobin synthesis inside the erythroblast, the structural integration process begins when two molecules of succinyl-CoA combine with two molecules of glycine to form:

a. A pyrrole ring
b. Protoporphyrin IX
c. A heme alpha chain
d. An iron core
e. An amino acid strand

74. Four individual pyrrole rings then combine to form a complex ring structure known as protoporphyrin IX, which subsequently binds with iron (Fe^{2+}) to produce a single molecule of:

a. Hemoglobin
b. Globin
c. Transferrin
d. Heme
e. Biliverdin

75. Each completed molecule of adult Hemoglobin A (HbA) contains a total protein component composed of four polypeptide globin chains, designated as:

a. Two alpha and two gamma chains
b. Two beta and two delta chains
c. Two alpha and two beta chains
d. Four beta chains purely

e. Four alpha chains exclusively

76. Because each Hemoglobin A molecule possesses four distinct heme groups, each molecule can bind reversibly with a maximum of how many individual atoms of oxygen?

a. Two atoms
b. Four atoms
c. Twelve atoms
d. One single atom
e. Eight atoms (forming four O_2 molecules)

77. In sickle cell anemia, a single genetic point mutation causes the substitution of valine for glutamic acid at the sixth position of the beta globin chain. When exposed to low oxygen concentrations, this variant hemoglobin (HbS) precipitates into long crystals that distort the red cell into a:

a. Perfect sphere
b. Biconvex star
c. Sickle shape (causing vascular occlusion and hemolysis)
d. Flat disc
e. Oval macrocyte

78. When old or fragile erythrocytes are trapped and broken down by the reticuloendothelial system (macrophages of the spleen and liver), the heme ring is cleaved to release iron and a green pigment that is quickly converted into:

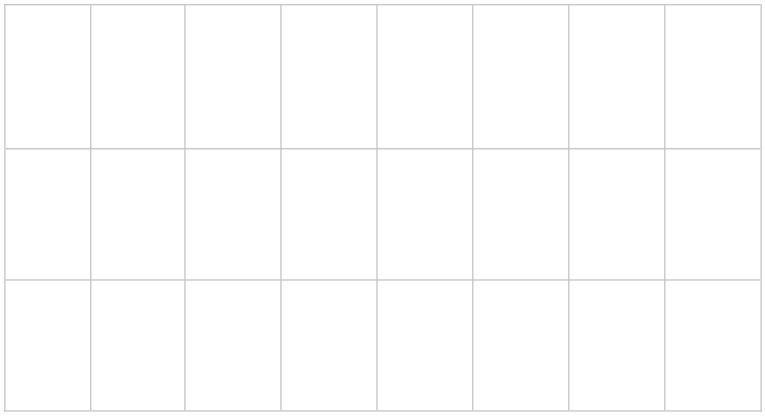
a. Biliverdin, then free bilirubin
b. Apoferritin
c. Transferrin complex
d. Urobilinogen purely
e. Stercobilin salt

79. The free, unconjugated bilirubin released into the plasma binds tightly to which transport protein to be carried safely to the liver for conjugation?

a. Haptoglobin
b. Transferrin
c. Fibrinogen
d. Alpha globulin
e. Plasma albumin

80. In the liver, hepatocytes detach bilirubin from albumin and conjugate it with glucuronic acid to form bilirubin glucuronide, which is then excreted via the bile ducts into the:

a. Urinary bladder
b. Intestinal tract
c. Pancreatic duct
d. Pericardial space
e. Hepatic vein



a	**b**	**c**	**e**	**b**	**e**	**e**	**e**